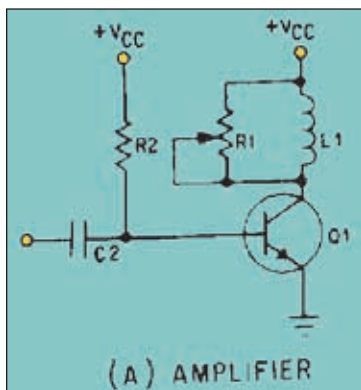


between the magnetic field of the inductor and the electric field of the capacitor.

2. The amplifier: as mentioned, charge oscillations in an inductor and capacitor alone would soon die down due to energy loss. Hence a device is required to keep amplifying the signal in order to cancel out the loss so that a constant sinusoidal signal may be maintained.

Such a device is usually realised by using a transistor to amplify the signals.



3. The feedback: the signal taken from the inductor-capacitor circuit needs to be fed back into the circuit after amplification. This is usually done through another inductor which realises this using the principle of electromagnetic induction.

Wire up the circuit as shown. C_b and C_e should be 30 microfarads while the variable capacitor box takes the position of C_1 . Inductors T_1 and L_1 are to be made by coiling wire around a pencil.

Instructions for wiring the transistor: of the three lines inside

